Synopsis For
(410248: Project Work Stage-I)

“Project Title”

SUBMITTED TO SAVITRIBAI PHULE PUNE UNIVERSITY

BACHELOR OF ENGINEERING

Computer Engineering

Submitted By

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Abstract (200–300 words)

- Problem domain & motivation
- Objective of the project
- Methodology (brief)
- Expected outcome/application

1) **Problem Statement**

- Define the problem clearly
- Specify scope and limitations

2) **Objectives**

   List **SMART** objectives (Specific, Measurable, Achievable, Relevant, Time-bound).

3) **Proposed System**

- Overview of the system
- Features & novelty compared to existing solutions
- High-level system block diagram

4) **Methodology / Approach**

- Stepwise process of project execution
- Algorithms/Models to be used
- Tools, frameworks, programming languages

5) **Expected Outcomes**

- Technical deliverables
- User/industry benefits

6) **Applications**

- Academic / Industrial / Societal relevance

7) **Future Scope**

- Scope for further development, extensions, research potential

8) **References**

- Follow **IEEE reference style** (Books, Research papers, Journals, Websites).

Guidelines

- Font: **Times New Roman, 12 pt, 1.5 spacing**
- Headings: 14 pt Bold
- Margins: 1 inch on all sides
- Page numbers at bottom center.

Sample

Abstract

With the growing use of medical imaging, accurate recognition and analysis of abdominal organs has become crucial in clinical diagnosis. This project, **AMORA – Abdominal Multi-Organ Recognition & Analysis**, aims to develop a deep learning–based system for automatic identification and segmentation of key abdominal organs (liver, kidney, spleen, pancreas) from CT and MRI scans. The system will assist radiologists in faster and more precise diagnosis, thereby reducing human error. The proposed methodology includes data collection, preprocessing, model training using convolutional neural networks (CNNs), and visualization of results. The expected outcome is a tool that can support healthcare professionals by improving diagnostic efficiency and accuracy.

1. Problem Statement

Medical image analysis is time-consuming and prone to human error. Manual segmentation of abdominal organs requires expertise and significant effort. There is a need for an **automated system** that can accurately identify and analyze multiple abdominal organs from imaging scans, ensuring faster and more reliable diagnosis.

2. Objectives

- To study existing medical image processing techniques.
- To collect and preprocess CT/MRI abdominal datasets.
- To develop a CNN-based model for organ recognition and segmentation.
- To evaluate the model using performance metrics such as accuracy, IoU, and F1-score.
- To design a visualization interface for healthcare professionals.

3. Proposed System

The proposed system uses **deep learning (CNN)** for automated organ recognition.

- Input: CT/MRI images
- Preprocessing: Noise removal, normalization
- Model: CNN/U-Net for segmentation
- Output: Highlighted regions of liver, kidney, spleen, pancreas with quantitative metrics

4. Methodology / Approach

1. Data Collection (Public medical imaging datasets like TCIA, CHAOS)
2. Data Preprocessing (Resizing, normalization, augmentation)
3. Model Development (CNN/U-Net architecture)
4. Training & Validation (80-20 dataset split)
5. Testing & Evaluation (Accuracy, IoU, F1-score)
6. Visualization (GUI using Python + Streamlit/Flask)

5. Expected Outcomes

- Automated segmentation of multiple abdominal organs from CT/MRI scans

- GUI-based tool for visualizing and analyzing medical images
- Reduced workload of radiologists
- Improved diagnosis accuracy and efficiency

6. Applications

- Hospitals and diagnostic centers for faster radiology analysis
- Academic research in medical imaging
- AI-driven healthcare systems

7. Future Scope

- Extension to 3D MRI scans
- Integration with hospital information systems (HIS)
- Deployment as a cloud-based diagnostic tool

8. References

[1] Smith J., et al. "Automatic Liver Segmentation Using U-Net." IEEE Access, 2020.
[2] Chen Y., et al. "Deep Learning for Multi-Organ Recognition." Medical Image Analysis, 2021.
[3] TCIA Public Datasets – https://www.cancerimagingarchive.net
[4] Goodfellow I., Bengio Y., Courville A. *Deep Learning*. MIT Press, 2016.